

Art and the science of generative AI

Ziv Epstein, Aaron Hertzmann, and the Investigators of Human Creativity

June 2023

Paper Review by Taha Cheema - 21204729

This paper discusses the opportunities and risks created by AI in the realm of art, including visual, auditory, and text forms. AI is an incoming disruption to art. Past disruptions include the advent of photography and digital music generation. But what is unique about AI is that it is trained on existing work and therefore, in a way, regurgitates original content.

An opportunity created by AI discussed in this paper is that of lowering the barrier to entry for creating art. This may support a wider range of creators which could boost the overall diversity of art products. I am inclined to not agree with this viewpoint because this lowered barrier to entry is enabled by generating content from trained data which is original work of other humans. Not only does this raise important concerns about ownership and copyright as the paper expands on, but also, to me, it seems to 'cheapen' art form. One can argue that humans also take inspiration from other artists' styles, so why is AI judged more harshly. I contend that there is still a creative aspect in the human generation scenario which is lost in AI because it is a model working on controlled denoising with attention on user prompts. Is it really original art if two individuals can write the same prompt and get similar results, with no creative touch of their own? Since they are not able to control the denoising explicitly at each stage, I feel that AI generated art forms may dilute the richness of human art. It is not that they will replace human art, but perhaps exist simultaneously, of which the pros and cons are numerous.

As far as risks go, the paper rightly points out concerns about homogeneity. Years from now, when AI art is widespread, future models could be trained on AI-generated art, leading to a self-referential "aesthetic flywheel". There are also concerns about biases in training data being amplified and economic effects as artists get displaced from certain creative roles; an example I can think of is social media design roles because of the recyclable nature of layout designs. Lastly, copyright and ownership of AI art remains a complex issue. These are real concerns and I agree with the paper's authors for highlighting them. I do think, however, that a very important concern of AI art that is only briefly mentioned in the paper towards the end is about misinformation via synthetic media, be it images or audio-visual content. This blurring of the boundary of real and AI could have a wide range of societal, social, and psychological effects that deserves greater debates (and guardrails). I think that the pace of AI development means that regulation and policy can sometimes lag behind, and that is a dangerous slope to slip on.